

Cell Design Specification — LNMO High-Voltage Smartphone Cell

Field	Value	Source
Electrochemical system	LNMO (LiNi0.5Mn1.5O4) high-voltage spinel cathode / graphite anode	`LNMO.json` positive override over `Chen2020` base
Nominal capacity (Ah)	4.5 (parameter set) / 6.372 (simulated 1C)	parameter set + `r5_d11_1c.json`
Voltage window (V)	2.5 – 4.7	parameter set (`Lower/Upper voltage cut-off`)
Cell dimensions (H×W×T)	electrode 65 mm × 1580 mm (unwound) × 204.6 μm total layer stack; shell thickness **Not provided**	parameter set + `r5_d11_energy.json:thickness_m`
Electrolyte formulation	high-conductivity: σ = 1.8 S/m (constant), ϵ_r = 0.4, D = Nyman2008 baseline (~1.77e-10 m²/s)	parameter bridge (formulation candidate)
Cation transference number	0.4	parameter bridge
SEI-suppressing coating	artificial SEI: SEI kinetic rate constant 2e-15 m/s (=500× reduction, **estimate**)	parameter bridge (coating candidate)

2. Electrode and Separator

Layer	Thickness (μm)	Porosity	Density (kg/m³)	Active fraction	Particle radius (μm)	Source
Positive (LNMO)	75.6	0.335	4400	0.665	5.22	parameter set (LNMO density, Chen2020 geometry)
Negative (graphite)	108	0.25	1657	0.75	1.5	parameter set + design override
Separator	7	0.47	397	—	—	design override (Chen2020 baseline 12 μm)
Positive current collector (Al)	8	—	2702	—	—	design override (baseline 16 μm)
Negative current collector (Cu)	6	—	8933	—	—	design override (baseline 12 μm)

- **N/P ratio**** ≈ 1.04 (negative areal capacity 64.4 Ah/m² ÷ positive 62.1 Ah/m²; negative = 0.895·Δx·c_max·L·ε_am, positive = simulated 6.372 Ah / 0.1027 m²).

3. Process Design Parameters

Parameter	Value	Formula / Source
Positive areal density	221.2 g/m²	75.6 μm × (1–0.335) × 4400 kg/m³
Negative areal density	134.2 g/m²	108 μm × (1–0.25) × 1657 kg/m³
Positive compaction density	2.93 g/cm³	4400 × 0.665 / 1000
Negative compaction density	1.24 g/cm³	1657 × 0.75 / 1000
Electrolyte fill amount	≈6.85 g	pore volume (5.71 cm³) × 1.2 g/cm³ (literature density, annotated)
Formation recommendation	0.1C CC to 4.7 V, 25 °C, 2 cycles	design-recommended value; production-line value requires tuning

4. Mass Breakdown (contract caliber, electrolyte excluded)

Layer	Mass (g/cell)	Source
Positive active (LNMO)	22.72	`r5_d11_energy.json:layer_kg_m2.positive_electrode` × area
Negative active (graphite)	13.78	`layer_kg_m2.negative_electrode` × area
Positive current collector (Al)	2.22	`layer_kg_m2.positive_cc` × area
Negative current collector (Cu)	5.52	`layer_kg_m2.negative_cc` × area
Separator	0.151	`layer_kg_m2.separator` × area
Total	**44.39**	`r5_d11_energy.json:mass_kg` (electrolyte/casing excluded)

5. Performance Verification (vs criteria entry-0)

Metric	Result	Threshold	Determination
1C capacity	6.372 Ah	—	—
Volumetric energy density	1260.2 Wh/L	≥ 950	✓
Voltage plateau (midpoint)	4.131 V	≥ 4.1	✓
4C charge T_{max}	321.68 K	$\leq 323.15 \text{ K}$	✓
4C plating	anode min +0.043 V $\rightarrow$ no plating	false	✓
SEI thickness after 100 cyc	193.6 nm	$\leq 500 \text{ nm}$	✓

6. Design Notes

- System switched to LNMO (4.7 V-class cathode) because the plateau $\geq 4.1 \text{ V}$ requirement excludes NMC811 ($\sim 3.6 \text{ V}$).
- Smartphone thin-cell architecture: current collectors thinned $16 \rightarrow 8 \text{ }\mu\text{m}$ (Al) / $12 \rightarrow 6 \text{ }\mu\text{m}$ (Cu) and separator $12 \rightarrow 7 \text{ }\mu\text{m}$, re-allocating the saved volume to a thicker anode ($85.2 \rightarrow 108 \text{ }\mu\text{m}$) — raises N/P (fixes plating) and volumetric ED ($1005 \rightarrow 1260 \text{ Wh/L}$) without increasing total stack thickness.
- Anode particles reduced $5.86 \rightarrow 1.5 \text{ }\mu\text{m}$ to cut 4C solid-diffusion polarization (τ from $\sim 1040 \text{ s}$ to $\sim 68 \text{ s}$).
- High-conductivity electrolyte ($\sigma 1.8 \text{ S/m}$, $t_{\text{ion}} 0.4$) reduces electrolyte concentration polarization.
- Artificial-SEI coating (SEI kinetic rate constant $2\text{e-}15 \text{ m/s}$) suppresses anode SEI growth ($753 \rightarrow 193.6 \text{ nm}$).
- Active cooling $h = 400 \text{ W/m}^2\cdot\text{K}$ keeps 4C-charge $T_{\text{max}} \leq 50 \text{ }^\circ\text{C}$ (321.68 K).

Honest caveats

- The 4C CC charge reaches $\sim 27 \%$ SOC before the 4.7 V cutoff: the positive electrode is diffusion-limited because the model uses Chen2020 NMC solid diffusivity ($4\text{e-}15 \text{ m}^2/\text{s}$) — the `LNMO.json` `Positive electrode diffusivity` override uses a deprecated parameter name and is not applied.
- Cooling $h = 400 \text{ W/m}^2\cdot\text{K}$ requires active thermal management (heat pipe / vapor chamber).
- SEI coating factor ($2\text{e-}15 \text{ m/s}$) is an estimate, not a simulated material.